Makers Making Change
A Neil Squire Program

Neil Squire

GAME Resources

None

None

None

Table of contents

1. Latest Updates

Version 1.1 | April, 2026 | Creating the site

Yay, this site exists! New format for the previous resources and a whole bunch of updated information

Version 1.0 | Jan, 2023 | Original version of this content

Launch of the original GAME Checkpoint Resources website where the resource guides and other materials lived

https://makersmakingchange.github.io/GAME-Checkpoints/

1.1 In Progress

A running list of resources we want to create. If you have time to help us work on these, please see the How to Contribute section and email us at info@makersmakingchange.com.

- **Player focused resource** - A static PDF that can be downloaded that give an abreviated version of the equipment/methods to make gaming accessible. Also will include a few example setups from other players and resources on where to find community
- **Expand Eye Tracking Resources** - Make more detailed resource on eye tracking under alternative access. Include info on using remote play on Xbox and PlayStation with systems on PC.
- **Expand Joystick/Switch Resources** - Play around with how to communicate the commercial and OpenAt versions of joysticks and switches.
- **Augmental MouthPad** - Now available. https://www.augmental.tech/
- **GazeStick** Using Eyegaze for right stick: https://crowking63.github.io/GazeStick/
- **Glydr** - foot controlled controller: https://glydr.gg/
- **Tell us how you are using these resources** - Add a like and review like page to get feedback and hearing from folks that are using the resources.
- **Audio Radar** - Audio Radar turns your game's directional sound into visual cues around your screen. Footsteps to your left, gunfire behind you, or movement ahead can activate the light bars in the direction the sound is coming from, giving Deaf and hard-of-hearing gamers access to information other players receive through audio.: https://www.youtube.com/watch?v=5iWhZ47dcik&feature=youtu.be
- **Add amazon list link from MMC of gaming gear:** https://a.co/d/0f9L83Ex
- **add video on what games we picked and rationale** - Provide a video and go over why we picked the games for checkpoints and discuss scaffolding.

2. Welcome to GAME Resources

2.1 What is this Resource?

Welcome! This resource is a collection of the Makers Making Change (https://www.makersmakingchange.com/), a program of Neil Squire (https://www.neilsquire.ca/), adaptive gaming resources. Intended for both or GAME Checkpoint community or anyone in the world interested in learning more about assistive technology in gaming. This resource assumes you are already familiar with the disability of someone else you are working with or you are reading this to learn about the assistive technology out there that you can use.

Link: Makers Making Change site

Link: Neil Squire site

2.2 Your Pathway to Play

This guide provides a structured approach to understanding the assistive technology in adaptive gaming, from initial hardware selection to real-world implementation.

2.2.1 Step 1: Adaptive Gaming Equipment

Understand the physical tools and software settings used to bridge the gap between a player and the game.

- Get an Overview of all the assistive technology options in gaming.
- Alternative Access: If standard controllers do not work for the player they could try using assistive switches, joysticks, voice control, facial expressions, eye control, and more.
- Controller Modifications: 3D printed and commercial modifications for a traditional controller
- Software Features: In-game accessibility and 3rd party applications for accessibility in games.

2.2.2 Step 2: Video Game Basics

Learn how to match a player's physical range of motion to the specific digital demands of a game.

- How to Find Games with Accessibility: Evaluating games based on the needs of the gamer.
- Video Game Literacy: New to gaming and don't know where to start?

2.2.3 Step 3: Best Practices

Recommended practices that we have developed or learned from other experts ensure a successful assessment and long-term setup.

- Session Walkthrough: A guide for walking through the before, during, and after an adaptive gaming session.
- Tips and Tricks: Recommendations that we often make for common requests such as one handed gaming, helpful considerations, and more.

2.2.4 Step 4: GAME Profiles

Review real-world case studies to see how all of the above sections are integrated for individual players.

- Gamer Profiles: Detailed equipment setups for various needs.

2.3 Additional Resources

Outside of the detailed resources above, we have a few additional tools for our GAME Checkpoints and pages on where to get support if you are an individual looking for gaming support or an organization looking to implement an adaptive gaming program.

GAME Checkpoint Resources
Access Questionnaires your clients can fill out before sessions, The base equipment list for a GAME Checkpoint centre, Marketing Materials, and Templates to help you document game accessibility and more.

Support
Find support. Look at organizations to help you on your adaptive gaming journey

Print These Resources
Download all of the adaptive gaming information in "Learn About Adaptive Gaming".

2.4 Why Adaptive Gaming Matters

Gaming is the largest entertainment industry in the world, providing a vital space for entertainment, careers, and community. However, accessibility challenges often remain a barrier to full participation.

For the 31% of gamers who live with a disability, adaptive gaming is life-changing:

- **Social Connection:** It reduces isolation by connecting players to a global community.
- **Independence:** It increases autonomy through digital participation as well as familiarty with assistive technology that can be used for digital access.
- **Rehabilitation:** It can be used to gamify physical and cognitive therapy sessions.

Makers Making Change works to ensure the disability community can fully participate by providing open-source assistive technology at about one-tenth the cost of similar commercial devices.

2.5 Makers Making Change - GAME Program Overview

Makers Making Change is leading Canada’s accessible gaming efforts. We’re supporting gamers, makerspaces, clinical centres, and game developers along their adaptive gaming journey. On this page you will find information around our low cost open source gaming assistive technology, resources on accessible gaming basics, R&D developments, and community initiatives.

The GAME Checkpoints program (https://www.makersmakingchange.com/game-checkpoint-program) is one of our main initiatives that was created o help organizations build accessible gaming spaces that fit their community. Many centres told us they wanted to support gamers with disabilities but needed guidance on equipment, setup, and training. Since 2022, we have worked with clinics, community hubs, schools, and game developers to create welcoming spaces where gamers with disabilities can explore adaptive gaming and find the equipment that works best for them.

Link: Visit the GAME Checkpoints Webpage

Scan to watch: Adaptive Gaming Overview Video

This resource is intended to be a resource guide and source of useful materials for the GAME Checkpoints as well as individuals looking for more information on adaptive gaming. If you are a GAME Checkpoint leader looking for specific tools, check out the GAME Checkpoint specific resouces on this site.

3. Learn About Adaptive Gaming

3.1 Adaptive Gaming Equipment

3.1.1 Equipment Overview and Platforms

How do I Know What I am Looking for?

There is plenty of assistive technology out there for adaptive gaming. Often times there is an overwhelming amount of options and it requires a combination of tools. This can be intimidating. This set of resources may seem overwhelming, but take the approach of determing what you are **NOT** interested in for yourself or the client you are working with. Put everything else on the "maybe" list and then begin evaluating the simplest option.

In general, there are three main ways to make adaptive gaming accessible:

1. Alternative Access
2. Controller Modifications
3. Software and In-Game Settings

Getting Started: Choosing Your Access Path

Whether you are a player looking for a personal setup or a clinician supporting someone else, the goal is the same: **maximize the player's existing access.** Use these two questions to decide which sections of this resource to explore first.

1. STANDARD CONTROLLER USE

Can the player use a standard controller, even if only partially (e.g., just one side or specific buttons)?

- **If Yes:** Start with Controller Modifications. You can often add 3D-printed parts or purchase controller modifications to make a standard controller more accessible.
- **If No:** Jump to Alternative Access. This shows you how to build a custom setup using tools like assistive switches, joysticks, and more.

2. SENSORY AND COGNITIVE NEEDS

Are there barriers related to vision, hearing, or how information is processed?

- **If Yes:** Explore Software Features first. There may be in-game accessibility or additional softwares that might make gaming more accessible for the player.
- **Note:** Sometimes using assistive switches or joysticks found in Alternative Access can also assisist cognitive needs by simplifying or enlarging the inputs to the game.

THE "ABILITIES-FIRST" STRATEGY

We don't focus on what a player *cannot* do. Instead, we identify every reliable movement a player **can** make and turn those into game inputs.

Case Study: The Hybrid Setup Consider a player who has full use of their left hand but limited movement on their right.

1. **The Base:** We might start with a Controller Modification like a one-handed controller.
2. **The Supplement:** If certain buttons are still hard to reach, we don't give up on that controller. We add a "Hybrid" element—perhaps a Assistive Switch or voice command—to handle the missing actions.

Pro Tip: Great setups are rarely one-size-fits-all. They are often a mix of hardware "hacks" and software settings.

Flow chart for Deciding What Route to Take

Before deciding on what three methods of making the game accessible you choose, you should consider both the game and gaming system/platform that you are going to be playing on. The assistive technology required to play a simple game on a phone is much different than playing a complex game on a PlayStation system.

The Platforms

You may already have a platform available to you or you may be looking for one to purchase. If purchasing a new platform, there may be some things you want to consider in terms of accessibility. **We are often asked, "What is the most accessible platform?" There is no single answer; while some systems offer more flexibility, the "best" choice depends entirely on individual needs and technical factors.**

We recommend starting with the equipment already available to you. Using what you have first is the most effective way to determine which features work for you and where you may need additional support.

COMPARISON OF CURRENT GENERATION PLATFORM ACCESSIBILITY FEATURES

Platform	Accessibility Considerations
Xbox Series X|S	• **Adaptive Controller** Xbox has their own "Xbox Adaptive Controller (XAC) that nativly connects to this system and Xbox One's. • **Controller Customization** Includes an app "Xbox Accessories" that allows you to remap and customize the inputs for an XAC or standard controllers. • **Built in Accessibility** check out the Xbox Accessibility Page (https://www.xbox.com/en-US/community/for-everyone/accessibility) for more information on the built in accessibility.
PlayStation 5 (PS5)	• **Controller Haptic Tuning:** Adjust vibration and trigger force on standard PS5 controllers. • **Adaptive Controller:** PlayStation has their own "Sony Access Controller (SAC)" that nativly connects to ONLY system. • **Built in Accessibility** check out the PlayStation Accessibility Page (https://www.playstation.com/en-ca/accessibility/) for more information on the built in accessibility.
PC (Windows)	• **Flexibility:** Compatibility for most controllers. • **How to get games:** You can use game launchers such as Steam, Epic Games, and more. Steam (https://help.steampowered.com/en/faqs/view/02F5-ACB2-6038-0F36) is the most common and has the most accessible options such as button remapping.
Nintendo Switch 2	• **Adaptive Controller** Nintendo has a 3rd party adaptive controller called the "Hori Flex Controller" that nativly connects to this system and Nintendo Switch 1's. • **Built in Accessibility** check out the Nintendo Switch 2 Accessibility Page (https://www.nintendo.com/en-ca/gaming-systems/switch-2/accessibility/?srsltid=AfmBOoqwke_w2a8pWIbG69hlJn1zSvBlcmMmyqBKNPsDJFwC0wgIg0Hc) for more information on the built in accessibility.
Mobile Gaming (tablets/ phones)	• **Adaptive Controller** The Xbox Adaptive Controller can connect to phones to allow assistive technology to work. There are also various switch interfaces made by assistive technology companies that can allow access. • **Built in Accessibility** the mobile device likley has various built in accessibility. This changes with each device/update frequently.

Link: Xbox Accessibility Page

Link: PlayStation Accessibility Page

Link: Steam Accessibility Page

Link: Nintendo Switch 2 Accessibility Page

COMPARISON OF PAST GENERATION PLATFORM ACCESSIBILITY FEATURES

Platform	Accessibility Considerations
Xbox Series One	• There are less accessible options built into this Xbox than the newer one, however, the Xbox Adaptive Controller is still compatible. • This system still had Controller Assist and all features through Xbox Accessories that are found on the Xbox Series Systems and is compatible with the XAC.
Xbox 360	• There are no adaptive controllers or specific alternative access for this platform.
PlayStation 4 (PS4)	• Note: The Sony Access Controller will not connect to this without an adapter.
PlayStation 3 (PS3)	• Note: The Sony Access Controller will not connect to this without an adapter.
Older PC's (Windows)	• It is important to consider the older and less powerful your PC is, the lower the ability to launch and successfuly play games is.
Nintendo Switch 1	• There are less accessible options built into this Nintendo Switch than the newer one, however, adaptive controllers like the Xbox Adaptive Controller (with an adapter) and Hori Flex Controller will still connect.
Nintendo Wii	• We often get asked about Nintendo Wii's because they were a popular device for rehabilitation centres. • There are no adaptive controllers or specific alternative access for the this platform.

STANDARD CONTROLLERS

When this resource uses the term "standard controllers" we are referring to the controllers that come default with the console or system. For example, a keyboard and mouse would be a standard input to a computer. Below are the main standard controllers currently still available by the platforms described above.

Xbox Controllers

XBOX CONTROLLERS

Controller	Visual Reference	Notes
Xbox Series X|S		• Looks very similar to the Xbox One controller, however there are slight differences of sizes and additional functionalities.
Xbox Series Elite 2		• This is the "pro" version of the Xbox Series X|S controller with added buttons on the bottom side of the controller and additional functionalities.
Xbox One		• This is the previous version of the Xbox Controller and is often mistaken for a Xbox Series X|S controller. • The easiest way to spot the difference is it does not have the "share" button right on the middle of the controller that the Series X|S controller does.

PlayStation Controllers

PLAYSTATION CONTROLLERS

Controller	Visual Reference	Notes
DualSense (PS5)		
DualSense Edge (PS5)		• This is the "pro" version of the DualSense controller.
DualShock 4 (PS4)		

Nintendo Controllers

NINTENDO CONTROLLERS

Controller	Visual Reference	Notes
Nintendo Joy-Con 2		
Nintendo Pro Controller 2		• This controller aims to look and feel more similar to what players expect from a standard controller. It looks similar to the Xbox controllers.
Nintendo Joy-Con 1		• Looks very similar to the Joy-Con 1's that were for the Nintendo Switch 1. There are key differences in size and connections.
Nintendo Switch 1 Pro Controller		• Looks very similar to the Nintendo Switch 1 Pro Controller. This version has buttons on the underside and other additional features.

It is important to note that the Nintendo Switch 1 and 2 Joy-Con controllers are designed in such a way that there is a lot of flexibility in how you can use them to play. There are two Joy-Cons (left and right or also called + and -) that work together or can work individually. There are three main types of ways you can hold/interact with the Joy-Cons.

> **Note**
>
> Some games will require you to play the whole game or certain parts in a specific mode. The Joy-Cons also have gyroscopic movement controlls in them that allow you to shake, wave, and move them around to interact. Again, only some games or part of games may utilize this feature.

Controller Mode This allows you to use both left and right Joy-Cons together to create a whole controller. The Nintendo Switch can be in its dock and connected to a TV or it can be out of the dock on its own stand.

Gamepad Mode Both Joy-Cons are attached to the Nintendo Switch directly and used out of the Nintendo Switch dock. This method can not be used in the dock.

Single Controller Mode Only one Joy-Con is used by the player. They hold it sideways and only have access to one thumbstick. Not every game can be played this way.

3.1.2 Alternative Access

Various MMC Assistive Switches and Joysticks

What is Alternative Access

Alternative Access referes to using any type of input into a game that **does not** involve using a "standard controller". In this case "standard controller" means the default controls that come with the system. This could be keyboard/mice or Xbox/Nintendo/ PlayStation controllers.

This section will provide an overview on all the devices and tools mentioned above as well as giving a quick overview of the platforms available.

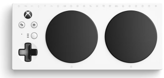
Adaptive Controllers

For the purpose of this resource, we are going to define adaptive controllers as anything built or designed for gaming platforms that either take a significantly different shape and/or allow assistive switch and/or joystick access.

There are three primary adaptive controllers out there with various approaches to making gaming more accessible. We like to divide them into **Hub Based Controllers** or **Direct-Use Controllers**. The difference is:

- **Hub Based Controllers:** Require other assistive technology to be plugged into it to create a setup. This is just the bridge between the assistive technology and the platform you are trying to game on.
- **Direct-Use Controllers:** Designed to be used without or with little addition of other assistive technology. The controller often takes a significantly different shape than the standard controller and has various joysticks and buttons built into it.

ADAPTIVE CONTROLLERS COMPARISON

Controller	Visual Reference	Approx Cost (CAD)	Category & Compatibility*	Key Features & Technical Details
Xbox Adaptive Controller (XAC) (https://www.xbox.com/en-CA/accessories/controllers/xbox-adaptive-controller)		$129.99	**Type:** Mostly Hub Based **Compatibility:** Natively compatible with Xbox One, Xbox Series X|S, computers, and phones/tablets.	**ow Detailed Info** • Hub based with enough buttons on the device out of the box to navigate the menu and play very simple games. • Most users will have to get additional assistive technology such as switches, joysticks, and mounting to create a custom setup. • You can plug in USB joysticks or joysticks that use 3.5 mm cables. Assistive switches can be plugged in as well if they have a 3.5 mm cable. • Uses the Xbox Accessories App (available on PC or Xbox) to remap and customize the inputs of the buttons. This allows it to be used as a gaming device or keyboard/mouse.
Sony Access Controller (SAC) (https://www.playstation.com/		$99.97	**Type:** Mostly Direct-Use **Compatibility:** ONLY compatible	

Controller	Visual Reference	Approx Cost (CAD)	Category & Compatibility*	Key Features & Technical Details
en-ca/accessories/access-controller/)			with the PS5 natively.	**ow Detailed Info** • Direct-Use controller with a built in joystick and circular array of buttons that can be remapped. • This also features 4 expansion ports that joysticks or switches with a 3.5 mm cable can plug into. No built in USB joystick compatibility. However, there are alternatives explained in the Sony Access Controller section. • Uses the settings inside of the PS5 to remap and customize the inputs of the buttons.
Hori Flex Controller (https://stores.horiusa.com/flex-controller-for-nintendo-switch/)		$440.00	**Type:** Hybrid (Hub/Direct-Use Mix) **Compatibility:** Connects to Nintendo Switch 1 and 2 natively.	

Controller	Visual Reference	Approx Cost (CAD)	Category & Compatibility*	Key Features & Technical Details
				ow Detailed Info • A mix between the Hub Based and Direct-Use controller approach. It has various 3.5 mm ports and two USB ports for joysticks (no 3.5 mm joystick access) but also an array of buttons on the top of the controller. • Has functionality to work with some eye tracking technology.

*Compatibility in this table is only referring to what the device was designed to connect with. However, you can use adapters to get these adaptive controllers to connect to nearly any platform. See the adapters section.

Link: Xbox Adaptive Controller (XAC)

Link: Sony Access Controller (SAC)

Link: Hori Flex Controller

See each section below for more specific information on the adaptive controllers.

XBOX ADAPTIVE CONTROLLER (XAC)

Released in 2018 with ongoing updates (even today), the Xbox Adaptive Controller (XAC) is a very powerful device with many applications. It could even be used as a bridge to use assistive technology to access a computer for personal use or employment. The XAC features some buttons on its face for usage out of the box but also has 3.5 mm and USB ports to add more assistive technology. Acting as a hub to create a custom adaptive gaming setup.

How to Get Started

Xbox has a fantastic series of videos and online resources explaining every feature on the XAC alongside player profiles where players using the XAC describe their setup.

Link: Scan to go to the XAC web resources created by Xbox

Check out the first video in Xbox's resource series here:

Scan to go to the Video XAC resources created by Xbox

SpecialEffect also has a fantastic walkthrough of setting up an XAC on an Xbox Console:

Scan to go to the SpecialEffect XAC Setup

Key Features

KEY FEATURES OF THE XBOX ADAPTIVE CONTROLLER (XAC)

Feature	What Does it Do	Use Case	Learn More
Assistive Switches and Joysticks	• The main feature of the XAC is being able to plug in 3.5 mm assistive switches and 3.5 mm or USB joysticks to create a custom setup	• If the built in buttons do not work for the player or they need additional input.	Found in: Video: XAC Guide - Getting Started (https://youtu.be/zd4VddU1wTQ?si=pKEhQ30Cf0V35zft&t=295)
Controller Remapping	• Each 3.5 mm port on the back of the XAC is labeled with a specific input. For example, you plug in an assistive switch to the A port, it will act as A. • However, you can use the Xbox Accessories App to reassign (remap) that button to any button or action available in the Xbox.	• You want to use the features such as toggle, shift, axis swap, etc. that are available in the Xbox Accessories App. You would set this up through the remapping process and save it.	Found in: Video: XAC Guide - Customization with the Xbox Accessories App (https://youtu.be/l73APLDxP1c?si=Nje4XIzBbI6t_i92&t=111)
Profile Saving	• You can save various remapping settings you have made for various games • You can save as many profiles as you want but you can only save 3 to the controller at a time to hotswap between. There is a button on the XAC to swap between the three profiles that can also be accessed by an assistive switch.	• You play a racing game, fighting game, and first person shooter. Instead of unplugging and replugging or remapping all the time, you can save the profile that works for that game once.	Found in: Video: XAC Guide - Customization with the Xbox Accessories App (https://youtu.be/l73APLDxP1c?si=mrOs_iFqMSdFT7lV&t=353)

Feature	What Does it Do	Use Case	Learn More
Controller Assist (previously co-pilot)	• This allows two controllers to act as one. Either two XAC's, standard controllers, or a standard controller and an XAC working together.	• One player does some of the inputs with the standard controller while the other uses the custom setup with the XAC to use those inputs.	Found in: Video: XAC Guide - Getting Started (https://youtu.be/zd4VddU1wTQ?si=pKEhQ30Cf0V35zft&t=295)
Button Toggle	• You can assign any button input to stay activated with a single press. Then press again to turn off. Think about a light switch.	• For users that do not want to hold a button down to keep it activated. Aiming in first person shooter games is a great example. Press once to aim, press again to put the weapon down.	Found in: Video: Gaming Readapted (https://youtu.be/peQryhh6aOw?si=3KyE7LBvb4FEY1G-&t=160)
Shift Mode	• Shift allows you to assign two different functions to a single joystick or button.	• You must choose an input (button or joystick) to make your shift button • The Shift key also allows you to assign two different functions to a single button. You can remap a button to be the A button A by default, but when holding the Shift key, your A button now functions as the B button when you press it.	Video: XAC Guide - Advanced functionalities (https://youtu.be/VuomjNhHYew?si=iLxLnji9b8kusAJx&t=41)
Joystick Axis Swap	• When remapping, if you select one of the joysticks it will allow you to swap	• This allows the gamer to play a game that requires 2 joysticks with one. • For example, if they were using a left	See the Tips and Tricks Section

Feature	What Does it Do	Use Case	Learn More
	either or both the X and Y axis.	joystick and swapped the X axis with the right joystick, they could move forward and back in a 3rd person game and use the left and right to look around and move in all directions.	
Joystick Sensitivty Curves	• Adjusting the sensitivity of the joystick plugged into the XAC.	• For example, players who have limited strength or mobility can choose a sensitivity curve option that provides an experience where less physical movement of the joystick is needed to achieve the same amount of in-game character or camera movement.	Video: XAC Guide - Advanced functionalities (https://youtu.be/VuomjNhHYew?si=NKKxagPeL2qmLz2Q&t=243)
Mounting	• 1/4-20 screw designed for AMPS compatible mounts. °-20 screw designed for tripod mounts.	• Placeing the controller in a more optimal position for the player	N/A

Link: Video - XAC Guide - Getting Started

Link: XAC Guide - Customization with the Xbox Accessories App

Link: Gaming Readapted - Button Toggle

Link: XAC Guide - Advanced functionalities

SONY ACCESS CONTROLLER (SAC)

Released in 2023, the Sony Access Controller took a different approach to the adaptive controller than both the XAC and Hori Flex Controller by attempting to create an out-of-the-box ready to play adaptive controller. This is a great option for some players but in other ways, its lack of customization in shape and limited amount of ports for additional assistive tech may not be enough for some.

How to Get Started

PlayStation has a fantastic series of videos and online resources explaining every feature on the SAC. The SAC also does a great job of walking the player through using the controller during first setup/launch on the system. This is by far the most intuitive adaptive controller.

Scan to go to the SAC web resources created by PlayStation

Check out the SAC setup and unboxing video:

Video: SAC Setup and Unboxing

Check out the SAC controller customization video:

Video: SAC Controller Customization

Key Features

KEY FEATURES OF THE SONY ACCESS CONTROLLER (SAC)

Feature	What Does it Do	Use Case	Learn More
Swappable Button Caps	• The SAC comes with various cap shapes (pillow, flat, curved, and overhang) that magnetically attach to the buttons. • Includes a "wide flat" cap that can cover two button sockets at once.	• A user needs a larger target area for a specific action. • Use the overhang caps for players who find it easier to pull a button rather than push it.	Button Caps Resource (https://www.playstation.com/en-ca/support/hardware/customize-access/)
Built-in Joystick	• A built-in analog stick that can have the cap swapped (standard, dome, or ball) and the arm length adjusted. • The orientation is 360° adjustable; you can set "North" to be any direction via the PS5 settings.	• A player is mounting the controller at an angle or upside down and needs the stick to still respond correctly to "Up/Down" movements. • Adjust the arm length to meet the player's reach.	Built-in Joystick Resource (https://www.playstation.com/en-ca/support/hardware/customize-access/)
Expansion Ports	• A total of four 3.5 mm expansion ports allowing for assistive switch or joystick (no USB) access • Up to four assistive switches or up to 2 joysticks	• If the built in joysticks or buttons do not work for the player or they need additional input.	Expansion Ports Resource (https://www.playstation.com/en-ca/support/hardware/access-expansion/)
Secondary Controller	• Allows up to three controllers to be paired as a single controller. This can include two Access controllers and one DualSense wireless controller. • All connected controllers can be used simultaneously to navigate menus and play games.	• A secondary user uses a DualSense controller to handle complex movements (like camera control) while the player uses one or two SACs for primary actions. • Two Access controllers are used together to create a full 360-degree button layout for a single player.	Secondary Controller Resource (https://www.playstation.com/en-ca/support/hardware/connect-multiple-access/)

Feature	What Does it Do	Use Case	Learn More
Controller Remapping	• Software-level customization that allows you to change what every physical button does on the PS5 system. • Allows for disabling buttons entirely to prevent accidental presses.	• You need to move a vital function (like R2) from a trigger to a large, accessible button on the SAC deck. • Simplify a game by removing unused inputs that might cause frustration.	How to set Access Controller Mapping (https://www.playstation.com/en-ca/support/hardware/access-profiles/#assign)
Profile Saving	• You can save various remapping settings you have made for various games. • You can save as many profiles as you want but you can only save 3 to the controller at a time to hotswap between. There is a button on the SAC to swap between the three profiles that can also be accessed by an assistive switch.	• You play a racing game, fighting game, and first person shooter. Instead of unplugging and replugging or remapping all the time, you can save the profile that works for that game once.	How to set Access Controller Mapping (https://www.playstation.com/en-ca/support/hardware/access-profiles/#assign)
Simultaneous Press	• Allows a single physical button to act as two button presses at the same time (e.g., L1 + R1). • The wide flat button cap can also be used to physically bridge two separate button sockets.	• Useful for fighting games or shooters where "Ultimate" abilities require pressing two buttons simultaneously. • Helps users with limited coordination who struggle to hit two separate targets at once.	How to set Access Controller Mapping (https://www.playstation.com/en-ca/support/hardware/access-profiles/#assign)
Button Toggle	• You can assign any button input to stay activated with a single press. Then press again to turn off. Think about a light switch.	• For users that do not want to hold a button down to keep it activated. Aiming in first person shooter games is a great example. Press once to	How to set Access Controller Mapping (https://www.playstation.com/en-ca/support/hardware/access-profiles/#assign)

Feature	What Does it Do	Use Case	Learn More
		aim, press again to put the weapon down.	
Joystick Sensitivity and Deadzone	• Adjusts how much physical movement is required to trigger an in-game action. • Deadzone settings allow you to tell the console to ignore small, accidental "drifts" or tremors.	• A player has very limited range of motion and needs the stick to be "highly sensitive" so a small twitch results in a full movement. • A player with tremors needs a larger "deadzone" so the camera doesn't shake.	How to set Access Controller Mapping (https://www.playstation.com/en-ca/support/hardware/access-profiles/#assign)
Mounting	• Two 10-24 screw holes for securely attaching the controller to mounts with an AMPS hole pattern. • One 1/4-20 hole located near the center of the controller to mount the controller to tripods or any mounts compatible with this screw hole.	• Placing the controller in a more optimal position for the player, such as on a wheelchair tray or a camera tripod.	Mounting Resource (https://www.playstation.com/en-ca/support/hardware/access-mount/)

Video: SAC Controller Customization

Video: SAC Expansion Ports

Video: SAC Secondary Controller

Video: SAC Controller Mounts

HORI FLEX CONTROLLER

The Hori Flex is a unique hybrid controller designed primarily for the Nintendo Switch (though it is also compatible with PC). It bridges the gap between the XAC and the SAC by offering a significant number of built-in buttons on the top of the device while still providing 3.5 mm ports for external switches and USB ports for joysticks. Notably, it is the only major adaptive controller that offers dedicated eye-tracking accessibility features right out of the box.

How to Get Started

Hori provided a detailed manual and others like SpecialEffect and Gaming Readapted have made fantastic video tutorials.

Link: Hori Flex Manual

Check out the SpecialEffect Hori Flex introduction and setup guide:

Video: SpecialEffect Hori Flex Setup

Gaming Readapted also provides a great overview of how the Hori Flex integrates with the Nintendo Switch:

Video: Gaming Readapted Hori Flex Setup

Key Features

KEY FEATURES OF THE HORI FLEX CONTROLLER

Feature	What Does it Do	Use Case	Learn More
Hybrid Input Design	• Features 18 assignable 3.5 mm switch ports and 2 USB ports for joysticks (no 3.5 mm). • Includes a small array of physical buttons on the face of the device for direct use.	• A user needs the "Hub" capability of the XAC but also wants a few physical buttons on the device for menu navigation or simple actions.	SpecialEffect - Hori Flex (https://www.youtube.com/embed/zak38eFZVm8?si=3XsJeCgceoZTgHNb)
Eye-Tracking Interface	• Features a dedicated interface that allows the controller to be operated via compatible eye-tracking devices on the PC.	• For players with very limited physical mobility who rely on eye-gaze technology to interact with their environment and games.	SpecialEffect - Hori Flex (https://www.youtube.com/embed/zak38eFZVm8?si=3XsJeCgceoZTgHNb)
Flex Controller Settings App	• A dedicated PC application used to remap buttons, adjust joystick sensitivity, and configure deadzones.	• You need to customize the "active" range of a joystick or change the behavior of a specific switch port for a Nintendo Switch game.	SpecialEffect - Hori Flex (https://www.youtube.com/embed/zak38eFZVm8?si=3XsJeCgceoZTgHNb)
On-Board Profile Storage	• Allows you to store multiple mapping configurations directly on the hardware. • Profiles can be toggled using a physical button on the controller.	• A user frequently switches between games like *Mario Kart* and *The Legend of Zelda* and needs instant access to different button layouts.	SpecialEffect - Hori Flex (https://www.youtube.com/embed/zak38eFZVm8?si=3XsJeCgceoZTgHNb)
Mounting	• Features a standard 1/4-20 threaded hole on the bottom of the device.	• Securely mounting the controller to a camera tripod or a magic arm for positioning near the head or lap.	SpecialEffect - Hori Flex (https://www.youtube.com/embed/zak38eFZVm8?si=3XsJeCgceoZTgHNb)

Video: SpecialEffect Hori Flex Setup

Cost Consideration

The Hori Flex controller is significantly more expensive than the other options. With the XAC providing very similar functions, compatibility with more devices (phones and computers), and a more dynamic remapping and customization software, consider purchasing an XAC with an adapter that allows it to connect to a Nintendo Switch 1 or 2.

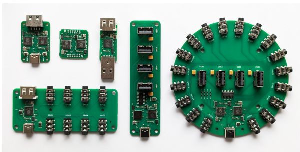
However, if the player wants eye tracking software compatibility specifically, that may shift the decision to the Hori Flex Controller.

OTHER ADAPTIVE CONTROLLERS

There are also a few other adaptive controllers out there. These typically only connect to phones and computers for gaming. This resource only touches on these briefly but see the table below for a few of the key options out there. The main reason someone may consider these options over the XAC, SAC, or Hori Flex is potentially due to reduced cost or customization available if they are open source and allow the user to customize the programming of the device.

OTHER ADAPTIVE CONTROLLERS COMPARISON

Controller	Visual Reference	Approx Cost (CAD)	Category & Compatibility*	Key Features Technica Details
Forest Hub (https://www.makersmakingchange.com/product/forest-joystick-mouse-hub/01tJR000000E4bdYAC)		~$75-150	**Type:** Hub Based **Compatibility:** Natively compatible with PC, tablets, and phones. Can be connected to XAC through left or right USB port.	• Enables user to connect a analog joystick (TRRS) a up to fou assistive switches emulate a USB Mou or USB Gamepad Another assistive switch ca used to c between and switc between Mouse an Gamepad mode.
Quester Switchbox (https://www.pretorianuk.com/quester-switchbox/)	Quester Switchbox	~$265.00	**Type:** Hub Based **Compatibility:** Natively compatible with computers and Android mobile devices. Works with Xbox via XAC USB ports.	• A "plug and-play" interface converts to six (6) assistive switches standard Android gamepad mouse in • Feature integrate display to show the current n and requ no extern

Controller	Visual Reference	Approx Cost (CAD)	Category & Compatibility*	Key Features Technica Details
				software configura making it ideal for clinical environm with restricted software installatio rights.
HID Remapper (https://www.remapper.org/)		N/A - Depends on the type of remapper and manufacturing costs	**Type:** Universal Adapter / Hub **Compatibility:** Broad; connects almost any USB input device to PC, consoles (via adapters), and mobile.	• A powe open-sou tool that allows fo hardware level remappin any USB device. It combine multiple controlle into one split one controlle across several outputs. • Suppor advanced logic like macros, layers, ar custom sensitivit curves st on the hardware bypassing need for

Controller	Visual Reference	Approx Cost (CAD)	Category & Compatibility*	Key Features Technica Details
				backgrou software.

*Compatibility in this table is only referring to what the device was designed to connect with. However, you can use adapters to get these adaptive controllers to connect to nearly any platform. See the adapters section.

Link: Forest Hub

Link: Quester

Link: Quester

Specialized Controllers

The distinction between Specialized Controllers and Adaptive Controllers is that adaptive controllers often act as a "hub" or intermediary between assistive technology and a platform. In contrast, specialized controllers are designed from the ground up with a specific form factor or function in mind. For example, to allow for one-handed use, modular building, or ultra-low-force interaction, etc.

BYOWAVE PROTEUS

The Proteus is a modular, "snap-and-play" controller system. It uses magnetic cubes and peripheral parts that allow a user to build a controller in whatever shape they need.

Link: Byowave Proteus

Controller	Key Features
Proteus Controller Kit	• Snap magnetic modules together to create unique shapes. • Akimbo Mode: A firmware update that allows two separate Proteus "Power Cubes" to work together as a single controller on one USB dongle. • Compatibility: Works natively with Xbox, PC, and Steam Deck.
Proteus Controller Builder	• Includes two of the cubes from the Proteus Kit and a 3D printed shell that is designed for one handed gaming.

NHUAD ONE-HANDED CONTROLLER

The Nhuad Controller is designed to place every function of a standard controller—including two analog sticks and all triggers for one handed gaming.

Link: Nhuad Controller

Controller	Key Features
Nhuad Controller	• One Hand Use: The design creates a layout to be fully operable with one hand. • Non Directional: Can be configured for either left or right-hand use. • Compatibility: Compatible with PC, PS4/*PS5*, and Xbox. (with adapters)

AZERON CONTROLLERS

Azeron devices are intended for one handed gaming. They have a variety of products that use keyboard and mouse input. Some of these units are adjustable as well. These are only compatible with systems compatible with mouse and keyboard inputs.

Link: Azeron Controller

Controller	Key Features
Cyborg 2	• Maximum Adjustability: The flagship model with 29 programmable keys and the most points of articulation for a perfect ergonomic fit.
Cyborg 2 Compact	• Low Profile: Features the same electronics and thumbstick as the Cyborg 2 but with a fixed, lower-profile button layout for users who prefer less vertical reach.
Keyzen	• Standard Keyboard Hybrid: Designed for those who want the Azeron ergonomics but in a more traditional keypad layout without the thumbstick.
Cyro	• Different Shape: A smaller and angled wrist placement with an array of buttons and a joystick for mouse/keyboard inputs. • Great for two joysticks: Also has a mouse sensor so the player can move it around on the table to get one joystick motion and use the thumbstick on it for the other.

8BITDO LITE SE/LITE SE 2.4G

The Lite SE was built specifically for gamers with limited mobility. It features a flat layout and low-resistance buttons so it can be played on a table or tray without needing to be held.

Xbox then collaborated with 8BitDo to make the Lite SE 2.4G. THis features compatibility with the Xbox and assistive switch ports.

Link: 8bitdo

Controller	Key Features
8BitDo Lite SE	• Low-Resistance: Joysticks and buttons are much easier to press than standard components. • Flat Face Layout: L3/R3 and all triggers are moved to the front face for easy access. • Compatibility: Natively supports Nintendo Switch, Android, and PC.
8BitDo Lite SE 2.4G	• Low-Resistance: Same as Lite SE • Flat Face Layout: Same as Lite SE. • Compatibility: Natively supports Xbox One, Xbox Series X|S, Android, and PC.

Assistive Tech for Adaptive Controllers

The adaptive controllers above all allow for some amount of additional assistive tech to be connected and used as an input in the game. These are often (but not limited to) assistive switches and joysticks. Think of these the same as the buttons and thumbsticks on a standard controller but coming in a variety of shapes, sizes, forces, and types of activation.

Finding assistive technology that works for the player but also the adpative controller/platform you are playing on is crucial. **Below are some resources to explain using assistive tech with adaptive controllers and where to find them.**

ASSISTIVE SWITCHES

Assistive switches act as the primary bridge between a user’s physical movement and a digital action. At their core, they are simply external buttons that can be mounted in a convenient location for the user, targeting whichever part of the body has the most reliable and comfortable range of motion. Most assistive switches utilize a standard 3.5 mm cable to connect to adaptive controllers or switch interfaces.

In a gaming context, these switches replace the physical buttons on a standard controller or keyboard that may be inaccessible to the player. By using a combination of these external "access points" a gamer can reliably and comfortably trigger any in-game input, from jumping to firing a weapon, using the movements that work best for them.

Here are the criteria that often seperates the assistive switch options out there:

- **Activation Force:** The amount of pressure (measured in grams) required to "click" the switch.
- **Target Shape:** The switch activation area can have a smooth or specific tactile surface, curve, or custom shape.
- **Target Size:** The surface area available to hit; larger targets assist gross motor movements, while smaller targets allow for precise mounting.
- **Travel Distance:** How far the switch must physically move before it activates.
- **Feedback:** Whether the switch provides a tactile "click" or an audible sound to confirm the button was pressed.
- **Mounting Type:** How the switch attaches to the environment (e.g., threaded inserts for arms or flat bases for Velcro).

Below are some links to various places you can get assistive switches. Depending on where you live, there may be more options available. This is just a basic list of common places we use to get you started to find options out there.

Open Source/DIY Options

These options have been released under an open source license. This means anyone should have access to the files to build and create this device.

ASSISTIVE SWITCH - OPENAT OPTIONS

Organization/ Device	Description	Link
Makers Making Change	• We host several assistive switches in our library with various toppers and mounting options. • Most of these switches use a 3.5 mm cable.	Search or filter the Assistive Devices for assistive switches (https://www.makersmakingchange.com/assistive-devices)
AbleGamers	• AbleGamers has a Printables page where they post designs	Link (https://www.printables.com/@AbleGamersCharity)
Online Repositories	• You may be able to find designs and files from those who have shared them on various online repositories for sharing open source designs. • Printables and MakerWorld are a good starting place to look.	Printables (https://www.printables.com/) MakerWorld (https://makerworld.com/en)

Link: Makers Making Change Assistive Devices

Link: AbleGamers Printables Page

Link: Printables

Link: Makerworld

Commercial Options

ASSISTIVE SWITCH - COMMERCIAL OPTIONS

Organization/ Device	Description	Link
Logitech Adaptive Gaming Kit	• A very reasonably priced set of assistive switches with hook and loop mounting trays and stickers for labeling the switches. • There is a version for the SAC and XAC, but they both will work interchangably. The only difference is the quantity of the trays and switches in the kit and type of sticker packs.	Link (https://www.logitechg.com/en-ca/shop/c/gamepads-controllers)
OneSwitch	• Various assistive switches.	Link (https://www.oneswitch.org.uk/shop.php)
Seven Mile Mountain	• Etsy page featuring various assistive switches.	Link (https://www.etsy.com/shop/SevenMileMountain?dd_referrer=https-3A-2F-2Fwww.google.com-2F)
Pretorian	• Various assistive switches.	Link (https://www.pretorianuk.com/)
HitClic	• Various assistive switches.	Link (https://hitclic.shop/en)
Canadian Assistive Technologies	• Canadian vendor for assistive tech.	Link (https://canasstech.com/)
Bridges Canada	• Canadian vendor for assistive tech.	Link (https://www.bridges-canada.com/)

Link: Logitech Adaptive Gaming Kit

Link: OneSwitch

Link: Seven Mile Mountain

Link: Pretorian

Link: Hitclic

Link: Canadian Assistive Tech

Link: Bridges

ASSISTIVE JOYSTICKS

Assistive joysticks provide a way to replicate the directional movement usually found on a standard controller's thumbsticks. While a standard joystick is fixed to the controller, an assistive joystick can be positioned independently to be used by the hand, foot, chin, or any other reliable point of movement. These devices translate physical movement into digital "X and Y" coordinates, allowing the player to move their character or control the camera in-game.

In gaming, assistive joysticks typically connect via a USB cable **OR** a 3.5 mm cable.

Here are the criteria that often separate the assistive joystick options out there:

- **Force Required:** The amount of physical strength needed to move the stick from the center position. Some "low force" sticks can be moved with a feather-touch, while others offer more resistance.
- **Range of Motion (Throw):** The distance the joystick must physically tilt to reach to provide input in the game.
- **Digital vs. Analog:**
- Analog: Responds to how far you push; move a little to walk, move a lot to run. Proportional control in all directions.
- Digital: Responds only to direction (like a D-pad); it is either "on" or "off.". Some work in 4 directions (up, down, left, right) and some work in 8-way (up, down, left, right, and the 4 diagonals)
- **Topper Style:** The physical shape of the handle (e.g., ball top, goalpost, or dome) which may be able to be swapped to match the user's prefered grip.
- **Input Connection:** Whether it uses a USB plug for direct-use/hubs or a 3.5 mm jack for specific adaptive ports.
- **Mounting Type:** How the base is secured, such as 1/4-20 threaded inserts for mounting arms, hook and loop, etc.

Below are some links to various places you can get assistive joysticks. Depending on where you live, there may be more options available. This is just a basic list of common places we use to get you started to find options out there.

Open Source/DIY Options

These options have been released under an open source license. This means anyone should have access to the files to build and create this device.

ASSISTIVE JOYSTICK - OPENAT OPTIONS

Organization/ Device	**Description**	**Link**
Makers Making Change	• We host several assistive joysticks in our library with various toppers and mounting options. • Some of these joysticks are USB and some are 3.5 mm (TRRS).	Search or filter the Assistive Devices for Joysticks (https://www.makersmakingchange.com/assistive-devices)
AbleGamers	• AbleGamers has a Printables page where they post designs	AbleGamers Printables Page (https://www.printables.com/@AbleGamersCharity)
Online Repositories	• You may be able to find designs and files from those who have shared them on various online repositories for sharing open source designs. • Printables and MakerWorld are a good starting place to look.	Printables (https://www.printables.com/) MakerWorld (https://makerworld.com/en)

Link: Makers Making Change Assistive Devices

Link: AbleGamers Printables Page

Link: Printables

Link: Makerworld

MOUTH JOYSTICKS

These are joysticks that are intended to be used with someones mouth or face. Most of the time, they include a sip and puff feature where people can breath in and out into the device to have inputs into the game.

There are two prominent options for mouth joysticks when it comes to gaming:

MOUTH JOYSTICK COMPARISON

Mouth Joystick	Approx Cost	Features	Considerations	Link
LipSync	$250	• **Light Force Joystick:** Requires very little physical pressure, making it ideal for users with limited head/ neck strength. • **Sip and Puff:** Single port system utilizing timed pressure (short, medium, long) for both sips and puffs to trigger distinct inputs. • **Interactive Hub:** Features 3 assistive switch ports, a built-in speaker for audio feedback, and a screen for visual settings. • **User Independence:** Users can access the Hub Menu independently to switch modes, calibrate, or adjust sensitivity without a PC. • **Multiple Modes:** Supports USB Mouse,	• A single sip/ puff port can limit the speed of inputs in fast-paced games where multiple simultaneous actions are required.	LipSync Page (https://www.makersmakingchange.com/product/lipsync/01tJR000000698fYAA) See it in Action (https://www.youtube.com/shorts/tC1QK64Vdmc)

Mouth Joystick	Approx Cost	Features	Considerations	Link
		Bluetooth Mouse, and USB Gamepad modes for versatile platform support. • **Adjustable Angle:** <br• **Compatibility:** Operates as a USB/Bluetooth mouse or USB Gamepad. It can plug into an Xbox Adaptive Controller (XAC) to bridge to consoles via adapters.		
Quadstick (FPS)	~$750 + fees	• **Sip and Puff:** Features four independent sip/puff sensors and a lip position sensor for complex, simultaneous commands. • **Custom Mapping:** Supports extensive macro profiles and "latching" inputs, allowing users to trigger complex button combos with a single breath. • **Compatibility:** Works natively	• Significant learning curve due to the complexity of breath-combination "codes." • Frequently out of stock due to high demand. • Mapping software has a learning curve. • There are a few different versions on the site.	Quadstick Page (https://www.quadstick.com/shop) See it in Action (https://www.youtube.com/watch?v=6CZqzKC1tdM)

Mouth Joystick	Approx Cost	Features	Considerations	Link
		with PC, PS3, PS4, and Nintendo Switch; compatible with Xbox and PS5 via adapters.		

Link: LipSync

Video: LipSync

Link: Quadstick

Video: Quadstick

Commercial Options

ASSISTIVE JOYSTICK - COMMERCIAL OPTIONS

Organization/ Device	Description	Link
Xbox Adaptive Joystick	• Developed by Xbox for the XAC. Compatible on its own to a computer, phone, or Xbox device.	Link (https://www.xbox.com/en-CA/accessories/controllers/xbox-adaptive-joystick)
Designed by Grier	• Various assistive joysticks.	Link (https://www.dbgrier.com/shop/assistive-technology)
Celtic Magic	• Products include the Feather Joystick (ultralight force) and Dangle Joystick (light force and low profile).	Link (https://www.celticmagic.org/)
OneSwitch	• Various assistive joysicks.	Link (https://www.oneswitch.org.uk/shop.php)
Seven Mile Mountain	• Etsy page featuring various assistive joysticks.	Link (https://www.etsy.com/shop/SevenMileMountain?dd_referrer=https-3A-2F-2Fwww.google.com-2F)
Pretorian	• Various assistive joysticks.	Link (https://www.pretorianuk.com/)
HitClic	• Various assistive joysticks.	Link (https://hitclic.shop/en)
Canadian Assistive Technologies	• Canadian vendor for assistive tech. They sell the Pretorian joysticks here too.	Link (https://canasstech.com/)
Bridges Canada	• Canadian vendor for assistive tech. They sell the Pretorian joysticks here too.	Link (https://www.bridges-canada.com/)

Link: Xbox Adaptive Joystick

Link: Designed by Grier

Link: OneSwitch

Link: Seven Mile Mountain

Link: Pretorian

Link: Hitclic

Link: Canadian Assistive Tech

Link: Bridges

Other Input Methods

If standard adaptive controllers, assistive switches, or specialized hardware do not meet all of a player's needs, consider integrating eye tracking, voice control, or gesture control. These "alternative inputs" can be used as standalone solutions or in combination with other AT to expand the total number of reliable inputs available to the player.

EYE TRACKING

Eye tracking technology allows a player to control a cursor or trigger game actions simply by looking at specific areas of the screen. While powerful, it often requires specialized hardware and software to translate "gaze" into the complex button presses required for modern gaming.

Check out this video from Adaptive Hacker Khan using a Tobii Eye Tracker as a game input using Millmouse and Project Iris

Video: Eye Tracking Tutorial

EYE TRACKING RESOURCES

Platform	Description	Link to Resource
PC / Computer	**Tobii Eye Trackers:** The hardware standard for eye tracking. For full game control, it is often paired with software like Millmouse or Project Iris to convert gaze into mouse movements and clicks (see resource).	Tobii Eye Trackers (https://gaming.tobii.com/product/eye-tracker-5/) Tutorial using eye tracker for full game access (https://youtu.be/bTFs6YgMtNg?si=yS9NrzxPh2plq900)
PC / Computer	**SpecialEffect Eye Gaze Games:** A collection of browser-based and downloadable games designed specifically to be played entirely via eye tracking.	Eye Gaze Games (https://www.eyegazegames.com/)
Consoles	**Hori Flex (Nintendo Switch):** The Hori Flex can be configured to accept eye-tracking inputs through specific PC-bridge setups. Refer to SpecialEffect resources for detailed wiring diagrams and "Eye Gaze to Switch" conversion methods.	SpecialEffect Hori Flex Guide (https://gameaccess.info/hori-flex-overview-video/)

Link: Tobii Eye Tracker

Link: Tobii Eye Tracker Tutorial

Link: Eye Gaze Games

Link: Hori Flex Eye Gaze Tutorial

VOICE AND GESTURE CONTROL

Voice and gesture recognition software translates spoken commands or physical movements (like a head tilt or a blink) into digital controller inputs. This is a rapidly evolving field that frequently uses AI to interpret a player's unique range of motion.

Check out this video from Cephable on using their software with Call of Duty: Black Ops 7 to use voice, button, and facial gesture input.

Video: Cephable Tutorial

VOICE AND GESTURE CONTROL RESOURCES

Tool	Description	Link
Cephable	**Multimodal Input:** An all-in-one platform that turns head movements, facial expressions, quick action buttons, and voice into inputs for PC and consoles. It features a highly customizable app to tune sensitivity and create macros.	Cephable (https://cephable.com/)
Playability	**Face & Voice Control:** Uses a standard webcam to map facial expressions and voice commands to play PC games and even PlayStation, Xbox & Switch via Remote Play or adapters. This allows a player to "smile" to jump or use voice cues to trigger inputs.	Playability Adaptive Software (https://playability.gg/)

Link: Cephable

Link: Playability

Important Considerations

The absolute crucial components of any alternative access setup are **mounting** and **adapters**. You can have the perfect selection of gear, but if you cannot securely position it where the player can reach it, the setup is practically useless. Likewise, if the hardware is perfect but doesn't talk to the console, it won't get the player into the game.

MOUNTING

Mounting is the process of securing controllers, switches, joysticks, or other assistive tech in a specific location that matches the player's reliable range of motion. Whether it’s on a wheelchair tray, a table, or a bed frame, the goal is "rock-solid" stability so the gear doesn't move during intense gameplay.

Below are a few common ways to approach this.

Hook and Loop

The most common starting point. Uses industrial-strength hook and loop to attach gear to lapboards or trays. It is highly adjustable and low-cost. You can start by taking a laptop tray, wheelchair tray, 3D printed surface and placing hook and loop fastener on it and the bottom of your assistive tech.

Articulating Arm

Arms that can be clamped to desks or wheelchairs. They allow for "3D" positioning, placing a the assistive tech exactly where it needs to be. There is a range of how sturdy these can be depending on the quality and design of the arm.

RAM Mounts

RAM mounts are modular ball-and-socket system known for extreme durability. AbleGamers created a series of adapters for common standard controllers to make them compatable with RAM mounts. It would also be possible have a flat plate at the end of the RAM attachment facing the user and put hook and loop fastener on it to attach assistive tech as well.

ADAPTERS

Adapters act as the "translator" between your standard controller, adaptive controller, or specalized controller to a specific platform they are not initally designed for. They are essential when using a controller or setup designed for one platform (like an Xbox Adaptive Controller) on a different one (like a PlayStation 5 or Nintendo Switch).

How to Find the Correct Adapter

Below are the main resources that our team uses to find an adapter that allows one controller to connect to a platform that it was not initially designed for.

RESOURCES TO FIND ADAPTERS

Tool	Description	Link
Gaming Readapted - Controller Connect Tool	A great resource for checking which adapters currently work with which consoles. This site is updated frequently as console firmware changes but may be slightly out of date.	Controller Connect Tool (https://www.gamingreadapted.com/controller-connect-tool)
Brook Gaming	Brook makes adapters and has several options. A good starting place.	Brook Website (https://www.brookaccessory.com/)
Brook FGC2	This is the adapter we use most often to get the XAC to work with the PS5.	Brook FGC2 Website (https://www.brookaccessory.com/products/wingmanfgc2/index.html)
Mayflash Magic NS2	This is the adapter we use most often to get the XAC to work with the Nintendo Switch 1 or 2.	Maylfash Magic NS2 (https://www.mayflash.com/product/magic_ns_2.html)
Cronus Zen	This adapter takes a different approach by using a "validating controller". It also has software to program things like macros. Also a good option to explore.	Cronus Zen Website (https://www.cronusmax.com/)

Link: Controller Connect Tool

Link: Brook Website

Link: Brook FGC2

Link: Mayflash NS2

Link: Cronus Zen

Adapters Go Out of Date

Console manufacturers often release firmware updates that can "break" third-party adapters. Always check the latest community forums before purchasing a specific adapter to ensure it still works with the latest system version.

3.1.3 Controller Modifications

Various Controller Mods

What are Controller Modifications?

Controller modifications involve altering a standard commercial controller to better suit a player's physical needs. This can range from "non-invasive" changes, like snapping on 3D printed extensions, to "invasive" hardware hacks where the controller is opened up to add new input ports. Whether you are adding a 3D printed grip, mounting the device for stability, or purchasing a professionally modded unit, these adjustments bridge the gap between standard hardware and a player's unique range of motion.

3D Printed Modifications

3D printing has revolutionized alternative access by allowing for low-cost, highly customizable physical interfaces. There is a vast library of open-source designs available that can be printed and attached to existing controllers without requiring any permanent changes to the electronics.

PLACES TO FIND CONTROLLER MODS

Device/ Organization	Description	Link
Makers Making Change	• Range of one handed, thumbstick, and other controller mods. • Can be requested from volunteer makers. • You can access the files to make yourself or request a device.	Makers Making Change Assistive Device Library (https://www.makersmakingchange.com/assstive-devices)
The Controller Project	• Supplying free controller modifications to gamers with disabilities or limb differences for many years. • You can access the files to make yourself or request a device.	The Controller Project Site (https://thecontrollerproject.com/)

Link: Makers Making Change Library

Link: The Controller Project

ONE HANDED CONTROLLERS

One-handed 3D printed adapters are among the most popular modifications. These mechanical rigs allow a player to access both analog sticks and all shoulder buttons using a single hand, often by resting the controller on a thigh or table to move the secondary stick via tilting.

Check out a one handed Mod in action:

Video: One Hand Mod in Action

Most one handed mods are made by a designer named Akaki. He has open sourced some of his designs but also sells them on his website directly. These cost approximatly $10 to 3D print at a local library or through our Makers Making Change program and $235+ if you buy from Akaki directly. If you want to buy them directly and not build them yourself or work with a volunteer, please view his website. Here is a list of current one handed modifications.

AKAKI ONE HANDED CONTROLLER MODS

Controller	Open Source Files	Akaki Controller Website
Xbox Series X|S	Open Source Files (https://www.makersmakingchange.com/product/onehanded-controller-modification-for-xbox-series-xs/01tJR000000690TYAQ)	Akaki Website (https://akaki.co/products/one-handed-xbox-series-xs-29039)
Xbox One	Open Source Files (https://www.makersmakingchange.com/product/onehanded-xbox-one-controller-modification/01tJR0000009St7YAE)	Akaki Website (https://akaki.co/products/one-handed-xbox-one-37695)
DualSense (PS5)	Open Source Files (https://www.makersmakingchange.com/product/onehanded-controller-modification-for-ps5/01tJR000000690SYAQ)	Akaki Website (https://akaki.co/products/one-handed-dualsense-70655)
DualShock 4 (PS4)	Open Source Files (https://www.makersmakingchange.com/product/onehanded-controller-modification-for-ps4/01tJR000000690MYAQ)	Akaki Website (https://akaki.co/products/one-handed-dualshock-4-15138)
Nintendo Switch 1	N/A - Has not open sourced these designs	Akaki Website (https://akaki.co/collections/products-for-nintendo-switch)
Nintendo Switch 1 • This is not an Akaki design	Open Source Files - Left Hand (https://www.makersmakingchange.com/product/lefthanded-grip-controller-modification-for-joycon/01tJR000000690RYAQ) Open Source Files - Right Hand (https://www.makersmakingchange.com/product/righthanded-grip-controller-modification-for-joycon/01tJR000000690VYAQ)	N/A

Link: Akaki Mods

Link: Xbox Series Mod

Link: Xbox One Mod

Link: DualSense PS5 Mod

Link: DualShock 4 PS4 Mod

Link: Nintendo Mod - Akaki

Link: Joy-Con Mod Left- Open Source

Link: Joy-Con Mod Right - Open Source

Adoption Rates

While powerful, one handed mods have a steep learning curve. Users often find they require significant practice to master the coordinated movements needed for modern games. For the players they work for, they work great.

THUMBSTICK TOPPERS

Thumbstick toppers are also a commonly requested controller mod. Xbox has even identified this and made their own website where a player can customize a thumbstick topper and get a 3D print file. Here are some common resources we use when looking for thumbstick toppers.

COLLECTION OF THUMBSTICK MODS

Thumbstick Topper Option	Description	Link
Xbox Adaptive Thumbstick Toppers	• Compatible with standard Xbox One, Xbox Series X|S, Elite, and Adaptive Joystick Controllers • Allows for customization of a range of toppers. Will give you a free 3D print file (.STL) that you can print them yourself or bring to a local 3D print vendor, possibly local library, or organization like Makers Making Change to print for you.	Xbox Adaptive Thumbstick Site (https://xboxdesignlab.xbox.com/en-ca/accessories/adaptive-thumbstick-toppers-for-xbox-controllers)
Thumb Soldiers	• Commercial thumbstick toppers for sale	Thumb Soldiers Site (https://thumbsoldiers.com/collections/accessibility/products/orbitalls-kit)
ActiveB1t Thumbsticks	• Open Source selection of 3D printed toppers for a range of controllers. • Download the files and print them yourself or bring to a local 3D print vendor, possibly local library, or organization like Makers Making Change to print for you.	ActiveB1t Printables Page (https://www.printables.com/model/1002274-thumbstick-topper-system-for-multiple-controllers/files)
AbleGamers Thumbstick Adapters	• Open Source selection of 3D printed toppers for a range of controllers. • Download the files and print them yourself or bring to a local 3D print vendor, possibly local library, or organization like Makers Making Change to print for you.	AbleGamers Printables Page (https://www.printables.com/model/989346-xbox-thumbstick-extension-system-accessibility)
Caleb Kraft Thumbstick Adapters	• Open Source selection of 3D printed toppers for a range of controllers. • Download the files and print them yourself or bring to a local 3D print vendor, possibly local library, or organization like Makers Making Change to print for you.	Caleb's Thingiverse Page (https://www.thingiverse.com/thing:658420)

Link: Xbox Adaptive Thumbsticks

Link: Thumb Soldiers

Link: Activeb1t

Link: AbleGamers Thumbstick Adapters

Link: Caleb Thumbstick Adapters

MODIFYING ADAPTIVE CONTROLLERS

It isn't just standard controllers that can be modded; adaptive hardware can also be customized. Here are a few modifications as an example:

- **Sony Access Controller (SAC):** Many makers have designed custom "toppers" for the stick and unique button shapes to help players with specific grip requirements. The team that made the Sony Access Controller made a "3D printing Guide" (https://www.playstation.com/en-ca/support/hardware/access-specifications/) with the specifications a designer would need to make custom joystick toppers and button caps.
- Designer, Harakan on Printables (https://www.printables.com/@Harakan_1552161) made various joystick and button cap toppers for the SAC.
- **Xbox Adaptive Controller (XAC):** Custom 3D printed knobs (like goals posts or large spheres) can be added to the joysticks used with the Xbox Adaptive Controller to accommodate different hand functions.
- Designer, Atom on Printables (https://www.printables.com/model/255246-d-pad-joystick-clip-for-xbox-adaptive-controller) made a joystick that can snap onto the XAC to modify the way a user would interact with the D-Pad.

Link: SAC 3D printing Guide

Link: Harakan Toppers

Link: Atom Printables

Commercial Modifications

For players who want a professional, "out-of-the-box" solution, several companies specialize in modifying controllers for accessibility.

- **Evil Controllers: (https://www.evilcontrollers.com/accessible-gaming)** They are a primary leader in this space, offering one-handed versions of the PS5, Xbox Series X, and Nintendo Switch controllers. These often feature re-routed buttons and analog sticks placed on the back of the controller for easier access.

Link: Evil Controllers

DIY Modifications

If a 3D printed part or commercial controller isn't a perfect fit, DIY materials offer a way to create a completely bespoke interface. Here is a common method we use.

- **Moldable Plastic (Instamorph):** This is a lightweight thermoplastic that becomes moldable in hot water and hardens into a strong plastic when cool. It is excellent for creating custom-molded finger grips, enlarging small buttons, or creating a personalized joystick topper that fits the exact contour of a player's hand.

Video: Moldable Plastic Tutorial

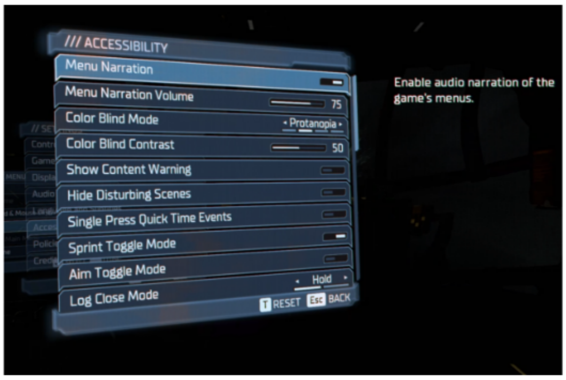
Mounting a Controller

A controller modification is only effective if the controller stays in the right place. Mounting provides the stability needed for players who cannot hold a controller's weight or who use 3D printed one-handed rigs.

- **RAM Mounts:** A modular ball-and-socket system that can securely hold a controller in any orientation. Many 3D printed mods include a "RAM Ball" base specifically for this purpose.
- **Hook and Loop (Velcro):** Industrial-strength Velcro can be used to secure a controller to a lap tray or desk, ensuring it doesn't slide away during intense sessions.

See the Mounting Section in Alt Access for more information.

3.1.4 Software and In-Game Settings



Accessibility Setting Menus in Games

What are In-Game Accessibility and Additional Software?

In-game accessibility refers to the settings and features built directly into a video game by developers. These allow players to customize the visual, auditory, and motor requirements of the game to match their specific needs.

Additional software refers to third-party applications that run alongside a game. These programs can "bridge" the gap when a game lacks certain features, such as adding voice control, eye-tracking overlays, or specialized screen reading capabilities.

In-Game Accessibility

Modern game developers are increasingly including robust accessibility menus. These settings can significantly reduce the physical or cognitive load required to play, often making the difference between a game being playable or unplayable.

To find resources on how to pick games with accessibility, visit How to Pick Games to learn more.

COMMON SETTINGS

Every game has a different set of accessibility features, and developers may use different names for the same settings or design choice they made to make it accessible to more players. While the **Accessible Games Initiative** **(https://accessiblegames.com/accessibility-tags/)** is working to standardize this language, it is important to recognize that settings are designed to address specific barriers across many different needs.

Link: Accessible Games Initiative

While many physical or motor settings are common, modern games also include features to support players with **low vision, blindness, colorblindness, hard of hearing or d/Deaf needs, cognitive or emotional barriers, and speech or strength and dexterity limitations.**

Below is a list of common accessibility features categorized by the barriers they help remove:

Motor, Strength, and Dexterity Settings

- **Input Remapping:** Allows players to rearrange button layouts to better suit their physical reach.

- **Toggle vs. Hold:** Converts actions requiring sustained pressure (like holding a trigger to aim) into a single press.
- **Sensitivity Adjustments:** Increases or decreases how much the game reacts to a joystick or mouse movement, helpful for players with limited range of motion or tremors.
- **Game Speed Adjustment:** Slows down the entire game to give players more time to react and process information.
- Also great for those with cognitive barriers.
- **Playable without Rapid Button Presses:** Removes "button mashing" requirements or Quick Time Events (QTEs).

Vision and Colorblind Settings

- **High Contrast Mode:** Simplifies the game's visuals, often highlighting characters and interactive objects in bright, solid colors against a dark background for **low vision or blind** players.
- **Screen Reader/Narration:**
- **Colorblind Filters:** Adjusts the game's color palette specifically for Protanopia, Deuteranopia, or Tritanopia to ensure critical information (like health bars or enemy teams) is distinguishable.
- **Text-to-Speech (TTS):** Narrates on-screen text, menus, and HUD elements for players with visual impairments.
- **Adjustable Text Size:** Increases the scale of subtitles and menu text for easier readability.

Hearing and Speech Settings

- **Subtitles and Captions:** Provides text for dialogue and, importantly, **directional captions** for environmental sounds (like footsteps or explosions) for **hard of hearing or d/Deaf** players.
- **Visual Cues:** Uses light or screen shakes to represent sounds that occur off-screen.
- **Speech-to-Text (STT):** Transcribes the voice chat of other players into text, allowing players to communicate without relying on audio.

Cognitive, Emotional, and Difficulty Settings

- **Difficulty Scaling:** Allows players to adjust enemy health, damage, or AI behavior to reduce the cognitive or reactionary load.
- **Game Speed Adjustment:** Slows down the entire game to give players more time to react and process information.
- **Wayfinding and Hints:** Provides clear visual paths or markers to the next objective to assist with navigation and memory.
- **Safe Modes:** Some games include settings to remove "emotional" triggers, such as disabling spiders (Arachnophobia mode) or reducing intense gore and flashing lights.

Here is a great talk from Alex Carey, founder of PlayAbility Consultancy, (https://www.play-ability.net/) showing examples of features for each of the above described barriers:

Link: PlayAbility Consultancy

Video: Alex Carey GAconf Talk

ACCESSIBILITY BY DESIGN

Some studios, like **Ubisoft**, lead the field by integrating accessibility into core design rather than as an afterthought. For example, in titles like *Far Cry* or *Assassin’s Creed*, they provide "Vision Packs" and "Motor Packs" that pre-configure groups of settings to help players get started quickly based on their specific disability profile.

These studios also think about accessibility outside of menus. For instance, a game might differentiate "Rare" vs. "Common" loot not just by color, but by unique shapes or icons so that a colorblind player can identify them at a glance without ever opening a settings menu.

Additional Software

When the built-in settings aren't enough, third-party software can provide a custom layer of control. These tools often translate one type of input (like a voice command) into something the game understands (like a keyboard press).

SCREEN READERS

For gamers who are blind or have low vision, screen reading software is essential.

- **Built-in Platform Readers:** Tools like **Narrator** (Windows), **VoiceOver** (iOS), and the native screen readers on Xbox and PlayStation can read out system menus and, in compatible games, in-game text and UI elements.
- **OCR Tools:** Some specialized software uses Optical Character Recognition (OCR) to "read" text appearing on the screen in real-time for games that do not natively support screen readers. Example from Ross Minor, using an OCR on Nintendo games (https://ablenews.com/blind-player-ross-minor-aims-to-make-video-games-accessible-for-all/).

Link: OCR on Nintendo Games

OTHER NOTABLE SOFTWARE

- **Voice Control, Eye Tracking, and Gesture Control:** These are all covered in detail in the Alternative Access Section under "Other Input Methods".
- **Remapping Suites:** Programs like **reWASD** (https://www.rewasd.com/) or the settings within **Steam** can provide much deeper remapping capabilities than standard game menus, allowing for "chorded" presses (triggering one action by pressing two buttons simultaneously) and advanced deadzone tuning for joysticks.

Link: reWASD

3.2 Video Game Basics

3.2.1 How to Pick Games

"What Games are Accessible?"

We get this question a lot, and the answer is almost always: **it depends.** Accessibility is highly personal. Whether a game works for you depends on several factors:

- What **genre** of game do you enjoy?
- What **gaming system** (PC, Console, Mobile) are you using?
- Do you want to play **online** with others or solo?
- What specific **accessibility settings** (visual, auditory, cognitive, or motor) do you require?
- How many **physical inputs** (switches, buttons, joysticks) can you reliably use?

Because this is such a personalized question, we recommend using dedicated resources to find the specific features you need. Find these below.

Resources for Finding Accessible Games

Rather than listing every game, we recommend these databases which are maintained by experts and disabled gamers:

- **AbleToPlay (https://abletoplay.gg/):** This site allows you to create a personalized profile based on your specific needs. It then generates customized game recommendations and "compatibility scores" to show how well a game might work for you.
- **CanIPlayThat (https://caniplaythat.com/):** A fantastic resource featuring deep-dive articles and reviews on game accessibility and hardware written by disabled gamers themselves.
- **Family Gaming Database (https://www.taminggaming.com/):** A massive database of board and video games. You can select the specific features in a game you are looking for (https://www.familygamingdatabase.com/en-us/search/show/controls+video+game) and filter or pick from a list of common features (https://www.familygamingdatabase.com/en-us/video-game-accessibility-data) to find games. This can help a ton when looking for games with specific accessibility features or games that “can be played with one switch”.
- They also allow you to filter by age, genre, etc. so this can be a great resource if you are specifically filtering for games for children.
- **Game Access (https://gameaccess.info/):** Create various articles revolving around how to use adaptive gaming hardware, showcase setups, and they also document accessibility in games. You can filter by "games" and find the articles related to specific game titles that explains its accessibility.

Link: AbleToPlay

Link: CanIPlayThat

Link: Family Gaming Database - Filtering tool, specific features

Link: Family Gaming Database - list of common features

General Guidance for Input Levels

Different genres of games require different levels of physical coordination and button complexity. If you are just starting your gaming journey or want games with limited controls needed, beginning with a "Low Input" genre can help reduce frustration while you get used to your assistive technology.

> **A Note on Experience**
>
> Choosing a game with lower input requirements does **not** mean sacrificing the quality of your experience. There is an incredible variety of art styles, deep storytelling, and immersive worlds within the "Low" and "Medium" input categories. Start where you feel comfortable and expand your library as you master your setup!

INPUT LEVEL DEFINITIONS

- **High:** Frequently requires two joysticks and multiple buttons that may need to be held simultaneously.
- **Medium:** Often requires two joysticks and several buttons.
- **Low:** Typically requires only one joystick and a limited number of buttons, or in some cases, 1-2 buttons only.

REACTION SPEED DEFINITIONS

- **High:** Requires rapid button presses, high-precision joystick movements, and immediate reflexes to avoid failure.
- **Medium:** Requires steady reactions to environmental changes, but usually provides a small window for error or predictable patterns.
- **Low:** Turn-based or slow-paced gameplay where the player has ample time to think and act without penalty.

GENRE COMPARISON

In general, genres often follow a similar playstyle. Therefore, to give some guidance, we broke down a list of popular genres and listed out their ussual input/reaction speed requirements:

GENERALIZED GENRE COMPARISON FOR INPUT AND REACTION SPEED

Genre	Input Level	Reaction Speed	Description	Examples
Puzzle	**Low**	**Low**	Requires solving logic problems or navigating cognitively geared challenges. Often turn-based with no time pressure.	*Portal, Candy Crush, Hidden Folks, Donut County*
Platformer	**Low**	**Medium**	Usually 2D movement (left/right) with jumping and basic actions to navigate levels.	*Super Mario Bros, Celeste, Cuphead*
Fighting	**Low**	**High**	Combat between characters, often on a 2D plane. Can be played against friends or the computer.	*Brawlhalla, Street Fighter, Dragon Ball Fighter Z*
Racing	**Low**	**Med-High**	Competitive or free-roam driving. Many offer "auto-drive" or simplified steering assists.	*Mario Kart, Need for Speed, Dirt 5*
Sandbox	**Medium**	**Low**	Players enter a world to customize, shape, and create their own path with little to no forced storyline.	*Minecraft, Terraria, Stardew Valley*
Sports	**Medium**	**Medium**	Simulates professional sports where you manage a team or play as an individual athlete.	*Nintendo Switch Sports, FIFA, NHL*
Action / Adventure	**High**	**High**	Story-driven games featuring exploration and combat. Requires complex camera and character movement.	*Spider-Man, Assassin's Creed, Far Cry*
	High	**High**	Fast-paced games played from a first-person	

Genre	Input Level	Reaction Speed	Description	Examples
Shooter / Battle Royale			perspective requiring high precision aiming and quick reactions.	*Call of Duty, Fortnite, Doom*
MOBA	**High**	**High**	Large-scale strategy games where teams work together to defeat the opposition in real-time.	*League of Legends, Dota 2, Smite*

Examples

SHADOW OF THE TOMB RAIDER (ADVENTURE)

This is a third-person action game where you navigate Lara Croft through dangerous environments. It requires the use of two joysticks (one for movement and one for the camera) and often requires precision aiming and shooting. While there are "stealth" and exploration sections that are slower, the intense combat and "Quick Time Event" sequences require almost every button on a standard controller to be used with very fast reaction times.

- **Input Level:** High
- Requires dual-analog stick coordination and simultaneous button presses for jumping/climbing.
- **Reaction Speed:** High
- Combat and environmental traps require immediate, precise responses to stay alive.

Check out gameplay here:

Video: Shadow of the Tomb Raider

MINECRAFT (SANDBOX)

Minecraft is a world-building game where you explore and create. To play fully, you do need two joysticks to move and look around, and you will use various buttons to build, mine, and manage your inventory. However, the game is largely taken at your own pace. By playing on "Peaceful" mode, you can remove all enemies, which eliminates the need for fast reaction speeds and allows for a stress-free creative experience.

- **Input Level:** High
- Requires two joysticks and multiple button inputs for inventory management and building.
- **Reaction Speed:** Low
- In creative or peaceful mode, there are no threats, allowing players to take as much time as needed for any action.

Check out gameplay here:

Video: Minecraft

DIRT 5 (RACING)

Dirt 5 is a rally racing game focused on off-road tracks. Racing games are excellent for alternative access because they can often be simplified; you primarily need one joystick to steer and two triggers (or buttons) for gas and brake. While the input count is low, the reaction speed is medium-to-high because the player must constantly react to turns, other drivers, and changing terrain to stay on the track.

- **Input Level:** Low
- Can be played with just one joystick and two primary buttons for acceleration/braking.
- **Reaction Speed:** Medium-High
- Requires steady reflexes to navigate corners and maintain speed against competitors.

Check out gameplay here:

Video: Dirt 5

Games with Simple Inputs

It is common for folks to be looking for games that use a single joystick with a single assistive switch or just simply less than 3 assistive switches. There are some great resources out there to find games with these inputs described in the Finding Accessible Games section. However, there are also some websites/resources that are dedicated to making one button, two button, single joystick and button games. Here is our go to sites for this:

GAMES WITH MINIMAL ASSISTIVE TECH REQUIRED

Resource	Input Needed	Cost ($CAD)	Description
AAC Gameplay (https://aacgameplay.com/)	1-2 switches, touch, mouse, or keyboard	Free	Collection of simple cause-and-effect and accessible games designed for AAC users and individuals with complex access needs.
OneSwitch (https://oneswitch.org.uk/art.php?id=28)	Single switch or keyboard key	Free	Directory of switch-accessible games and activities that can be controlled with one input.
Padlet: Arcade Switch Games (https://padlet.com/loreto5/arcade-switch-games-asbb646i7xsq5cfs)	Single switch or keyboard key	Free	Curated collection of browser-based games suitable for switch users and beginners.
EasyToPlayGames (https://easytoplaygames.itch.io/)	1-2 buttons, keyboard, or touch	Free	Simple browser games with minimal controls and immediate visual feedback.
T-Rex Runner (https://trex-runner.com/)	Single button or switch	Free	Endless runner game where a single input makes the dinosaur jump over obstacles.
HelpKidzLearn (https://www.helpkidzlearn.com/)	Single switch, touch, mouse, or eye gaze	Subscription	Educational and cause-and-effect games designed for learners with disabilities.
SpecialBites (https://specialbites.com/)	Single switch, touch, mouse, or eye gaze	Subscription	Cause-and-effect activities and games developed for switch users and emerging communicators.
Shiny Learning (https://shinylearning.co.uk/)	Single switch, touch, mouse, or eye gaze	Subscription	Interactive learning activities and games with accessible input options.
Papunet Game Pages (https://papunet.net/gamepages/games/)	Single switch, keyboard, mouse, or touch	Free	Accessible online games and activities designed for users with diverse abilities.

Resource	Input Needed	Cost ($CAD)	Description
NARBE House (https://narbehouse.github.io/bennyshub/)	Single switch, keyboard, or mouse	Free	Collection of simple browser-based switch-accessible games and cause-and-effect activities.
Ginger Tiger (https://gingertiger.net/)	Single switch, keyboard, touch, or eye gaze	Free / Subscription	Accessible games and activities focused on engagement, communication, and skill development.
Otto Ojala Games (https://ottoojala.itch.io/)	1-2 buttons, keyboard, or gamepad	Free	Minimal-input browser games with simple interactions and clear visual feedback.

Make your own games with simple inputs using Microsoft MakeCode Arcade (https://www.microsoft.com/en-ca/makecode/teach/).

3.2.2 Video Game Literacy

What is Video Game Literacy?

In the world of assistive technology, we often talk about **digital literacy**—the ability to find, evaluate, and communicate information through various digital platforms. It involves understanding how to navigate a computer, use a mouse, and manage files.

Video game literacy applies these same concepts to the gaming world. Someone who grew up gaming might instinctively know that the bottom face button (A on Xbox or X on PlayStation) is usually "Jump," while the right face button (B or O) is usually "Go Back" or "Cancel." However, for a clinician or a new gamer, these "unwritten rules" are not always obvious. Literacy involves understanding how to navigate in-game menus, interpreting visual icons (like a health bar or a quest marker), and knowing the general "language" of how games behave.

To see a great breakdown of how these invisible rules affect new players, watch this video:

Video: Video game literacy

Game Modes

Inside a single game title, there may be various modes that change the experience slightly or entirely. Take **Fortnite** for example: the base game is a battle royale shooter, but it now includes modes for racing, rhythm games, and LEGO-style survival—all within the same app.

While some games only offer a single way to play, most modern titles include a few of the following:

- **Single-Player / Campaign / Story:** This is the core experience where the gamer follows a "main" narrative. It is usually played solo and allows the player to progress at their own pace.
- **Training Mode / Freeplay:** These are low-stakes environments where gamers can practice movement and mechanics without the pressure of enemies or a timer. These are excellent for clinicians to use when first testing a new adaptive setup.
- **Custom Game:** Allows the player to set their own rules, such as turning off specific hazards, extending time limits, or playing on specific maps.
- **Multiplayer:** This allows an individual to play with or against others. This can be "Local" (sitting on the same couch) or "Online" (via the internet).
- **Co-op:** Players work together to achieve a goal.
- **Versus:** Players compete against one another.

Game Compatibility

Understanding how games connect across different devices is a key part of game literacy, especially when trying to play with friends or family.

- **Crossplay:** This refers to a game that allows players on different platforms to play together. For example, a person on an Xbox can join a match with a friend on a PlayStation or a PC.
- *Note:* It is important to check if the game supports cross-generation play (e.g., a PS4 player playing with a PS5 player).
- **Cross-platform:** This simply means a game is *available* on multiple platforms (like both Xbox and PC). It does **not** always mean those players can play together (Crossplay). Always check for the "Crossplay" tag if the goal is social gaming across different systems.

3.3 Best Practices

3.3.1 GAME Session Walkthrough

Gaming with assistive technology is a personal journey. There is no one-size-fits-all solution; instead, these best practices serve as a roadmap based on Makers Making Change's experience working with partners such as the Stan Cassidy Centre for Rehabilitation. This is intended to help a clinician, family member, or player think through trialing adaptive gaming equipment.

The Gaming Session

Trialing adaptive gear before purchase is important for finding a setup that works. While every center operates differently, the following framework provides a flexible guide.

GENERAL SESSION BREAKDOWN

GAME Session Step	Why	Tool/Resource
1. Before	Gather user information and prepare initial solutions	Gamer Questionnaire
2. During	Test and adjust equipment and settings	Equipment Sections
3. Assess	Evaluate outcomes and determine next steps	Equipment Sections
4. Finalize	Document and support setup acquisition	Support Options

⚠ **These Steps are Flexible.**

Customize this workflow based on your resources and goals.

1. GAME Session - Before

Question: Has a clear gamer goal been identified?

Yes

Focus the session on that goal (specific game, activity, or experience).

No

- Ask what games or genres interest them
- Ask if they have played before
- Ask what they want to achieve

If unsure, start with a simple, low-input game.

Question: Is the preferred gaming system known and available?

Yes

Use their preferred or most-used system.

No

- Ask what system they want to use
- Consider where their friends play
- Consider types of games

If unavailable:

- Try a similar system
- Use adapters where possible
- Choose the best available system for the trial

Question: Has an appropriate trial game been selected?

Yes

- Matches their goal or genre
- Has manageable input requirements
- Has a low-stress entry point (tutorial, freeplay)

No

- Match the genre or goal game
- Start with fewer inputs if needed
- Choose slower-paced or simpler games

If needed:

- Start simple and progress toward the goal game

Question: Has an appropriate controller or access method been selected?

Yes

- Standard or modified controllers
- Switch interface setups
- Alternative inputs (e.g., mouth joystick)

No

- Try a standard controller to assess barriers
- Try a switch interface if needed
- Consider mounting or alternative layouts

Question: Is the setup prepared and stable?

Yes

- Equipment is available
- Game is ready to launch
- Mounting solutions are stable

No

- Prepare mounts (RAM, trays, Velcro)
- Gather controllers and adapters
- Check game updates and starting point

Initial Game Selection

- You do not have to start with their gaming goal game
- Try a low risk situation to make sure everything works
- We often use the game Brawlhalla under training mode. Only requires one joystick and a few buttons, free, and fun!
- If there is a large gap between the user's current gaming abilities or number of inputs they can access, and the number of inputs that their goal game will require consider starting not with their gaming goal game.

2. GAME Session - During

Question: Can the user begin interacting with the game?

Yes

Move into gameplay and observe performance.

No

Identify the issue:

- Navigation difficulty
- Physical access issue

Adjust:

- Button mapping
- Positioning
- Provide guidance if needed

Question: Are barriers present during gameplay?

Yes

Game-related barriers:

- Adjust difficulty or settings
- Try different modes
- Try a different game if needed

Controller-related barriers:

- Change layout or mapping
- Try different hardware
- Adjust switches or joysticks
- Consider mounting or alternative inputs

No

Continue gameplay and confirm consistency

Question: Is the user experiencing fatigue or frustration?

Yes

- Pause the session
- Reduce effort required (force, positioning)
- Take breaks

No

Continue while monitoring comfort

Question: Has the setup been documented?

Yes

- Equipment used
- Layout and positioning
- Game settings

No

- Record what worked and what didn't
- Take photos of setup
- Note barriers and solutions

3. GAME Session - Assess

Question: Did the setup meet the gamer's goal?

Yes

Move forward with finalizing the setup.

No

- Schedule another session if possible
- Identify additional equipment to try
- Document remaining barriers

Question: Are additional resources or equipment required?

Yes

• Explore new hardware options
• Try different game approaches
• Look into external supports

No

Proceed to finalize

Question: Does the user need external support?

Yes

• Makers Making Change
• AbleGamers
• Funding options if available

No

Continue to final setup

---

4. GAME Session - Finalize

Question: Has a full equipment list been created?

Yes

• Full list of devices
• Required accessories
• Purchase links if possible

No

• Document all hardware used
• Include setup notes
• Consider home setup requirements

Question: Does the setup include open-source devices?

Yes

- Submit a request through Makers Making Change
- Include device details
- Attach notes and photos

No

Proceed with commercial options

Question: Has the user received all setup information?

Yes

- Confirm understanding of setup
- Confirm settings and equipment sourcing

No

- Provide equipment list
- Provide setup photos
- Provide settings and instructions

Question: Is additional support required?

Yes

- Schedule follow-up session
- Provide setup guide with images
- Assist with equipment delivery if possible

No

Ensure the user can independently set up and use the system

3.3.2 Tips and Tricks

There are many small decisions that can make or break an adaptive gaming setup. Over time, players, clinicians, and assistive technology specialists have developed practical strategies that improve access, comfort, and overall experience.

This section captures a mix of proven approaches and commonly shared tips. Treat these as flexible ideas—what works for one player may not work for another.

Player Independence

Do not overlook system and menu navigation. Being able to independently start, pause, and navigate a game can significantly increase a player's sense of control and confidence. Here are some common things to consider when thinking about a players independance.

- Ensure the player can:
- Turn the console on/off
- Launch a game
- Pause or exit gameplay
- Place "non-critical" actions (menu, pause, home) on less "easy to activate" inputs if needed. Prioritize the reaction speed for inputs more relevant to the game.
- Use features like **Shift Mode (XAC)** to add secondary functions without increasing the amount of hardware or inputs. Think, a switch could have the shift function to be "switch profiles" but A in its primary mapping.
- Consider accessibility features built into the system (e.g., remapping, shortcuts)

Gamer Experience

Fatigue, frustration, and enjoyment matter just as much as technical success.

Gaming should not feel like a workout (unless that is the rehabilitaiton goal) or cause pain.

Tips:

- Monitor for:
- Fatigue
- Strain
- Frustration
- Take breaks early—don't wait until the player is exhausted
- Reduce force requirements where possible (lighter switches, lower joystick tension)
- Consider alternative mounting locations for the assistive tech to reduce strain.

Important mindset:

The experience the gamer wants does not need to match their current ability.

- Start with the **desired experience**, not just what is easiest to create a setup for.
- If a player wants to play a more complex game:
- Build toward it
- Use assistive tech to expand access
- Avoid forcing players into games just because the game is "simpler"

For example, if someones gaming goal is to play Call of Duty and they only like first person shooter games, creating a setup that only works for pinball may be a great start, but that is not where the journey should end.

Joystick Related

Tips and strategies when working with joysticks.

ACCESSING TWO JOYSTICKS FROM A SINGLE JOYSTICK

Many games require two joysticks, but not all players can physically access both.

There are ways to extend a **single joystick** to cover more functions.

Here are Three Tricks for This:

Check out the video detailing the three methods:

Video: Convert 2 Joysticks Into 1 Joystick

1. **X Axis Switching (XAC)**
- Using the XAC and Xbox Accessories App, swap the X axis on the left joystick with the right joystick
- This will allow you to move forwards and backwards while using left and right to control the direction of the character.

2. **Shift Mode (XAC)**
- Use the primary mode for one joystick and the shift mode for the other.
- Use an assistive switch in either toggle or regular shift mode to swap between using the left or right joystick.

3. **“Walk Forward” Button**
- Map forward movement to a button
- Use joystick for camera movement to control direction (typically right joystick)
- **Can sometimes be done inGame Settings**
- Look for:
- Auto-run
- Camera assist
- Reduced need for dual-stick control

SpecialEffect has created a fantastic resource for this:

Video: Special Effect - walk forward

USING A MOUTH JOYSTICK

A mouth joystick can be a powerful addition to a setup.

Benefits:

- Adds an additional joystick input
- If you have switch or joystick access below your neck, it frees more access to:
- Buttons
- Switches
- A second joystick
- Can reduce complexity of hand-based inputs

Use cases:

- Players with limited hand mobility
- Players who need access to a second joystick but have no more inputs available below their neck.
- Players with no movement below their neck.

MOUNTING ASSISTIVE SWITCHES ON JOYSTICKS

Switches can be mounted directly onto joysticks or toppers to enable quick access actions.

Methods:

- Hook and loop fastner
- Moldable plastic
- Custom 3D printed mounts

Benefits:

- Faster activation for frequent actions
- Reduced reach distance
- Better integration of controls

Example:

Utilizing Shift Mode on the Xbox Adaptive Controller (XAC)

Shift Mode allows a single input to perform multiple functions depending on whether a “shift” button is held.

Benefits:

- Doubles the number of available inputs without adding hardware
- Reduces physical reach requirements
- Allows layering of controls (primary vs secondary actions)

Common uses:

- Switching between movement and camera control on a joystick
- Adding menu/navigation functions
- Expanding limited switch setups. You can effectivly double your inputs by adding a shift mode to them.

Considerations:

- Ensure the player understands when Shift is active
- Avoid overly complex mappings
- Test for cognitive load and usability

Using the Xbox Adaptive Controller on the Nintendo Switch 1 or 2

HOW TO CONNECT

We typically use the **Mayflash Magic-NS 2** (https://www.mayflash.com/product/magic_ns_2.html) adapter to get the Xbox Adaptive Controller (XAC) working on the Nintendo Switch. See more in the Alternative Access section.

Link: Mayflash

For instructions on connecting the XAC to the Nintendo Switch (it is the same on the Nintendo Switch 2), please seee this video below.

Video: How to setup XAC on Nintendo Switch

HOW TO REMAP THE CONTROLS OF THE XAC

There are two main ways to remap

1. In the Nintendo Switch settings

- In the settings of the Nintendo Switch, you will find the option to change the button mapping in the controller settings (https://en-americas-support.nintendo.com/app/answers/detail/a_id/49229/~/how-to-change-the-button-mapping-on-nintendo-switch-controllers)

2. In the Xbox Accessories App using PC or Xbox

- If you create a profile in the Xbox Accessories App it will be saved to the XAC itself. So when you go back and use it with the Nintendo Switch, the up to three saved profiles will be available.
- See more about this in the Xbox Adaptive Controller Section

Link: Nintendo Controller Settings

USING 2 CONTROLLERS AS ONE (CONTROLLER ASSIST) ON NINTENDO SWITCH 1|2

The Nintendo Switch 1|2 do not have a direct "controller assist" or "assist controller" mode where you can use two or more controllers as one. However, there is a method to do this using the Mayflash NS 2 adapter. This would allow you to:

- Combine:
- XAC + Nintendo Pro Controller
- XAC + single Joy-Con
- XAC + double Joy-Con
- Joy-Con + Nintendo Pro Controller

Instructions:

See the video tutorial below to see how to set this up:

Video: How to Copilot on Switch

WHY ARE THE BUTTONS SWAPPED?

Button layouts differ between systems, which can cause confusion. For example, when you press A, B, X, or Y buttons on the XAC or they do not directly correlate to the A, B, X, and Y buttons on the Nintendo Switch. This is because, on the standard controller, the buttons are placed in **differnt locations**. See the photo below.

Therefore:

XBOX BUTTONS AND WHAT THEY DO WHEN USED ON THE NINTENDO SWITCH

Xbox	Nintendo Switch
A	B
B	A
Y	X
X	Y

Quick fix:

- Hold **Pause (menu button) + A for ~3 seconds** (adapter dependent) on the XAC
- This swaps inputs to match expected layout.

Tip:

Always test button mapping before starting gameplay. You can do this through the test input devices (https://en-americas-support.nintendo.com/app/answers/detail/a_id/22451/~/how-to-test-the-controller-buttons-on-nintendo-switch) on the Nintendo Switch.

Link: Test Inputs

Adapters and the Button Layout Problem

Adapters can introduce confusion in button mapping. For example, if you are using an Xbox controller like the XAC on the PlayStation 5, there is no Square on the Xbox controller, so which assistive switch port on the XAC would you plug an assistive switch into to use as square on the PS5? This is what this tip will help with.

Different systems use different layouts:

- Xbox: A, B, X, Y
- Nintendo: A, B , X, Y (but in a different location on the controller)
- PlayStation: Cross, Circle, Square, Triangle

Key Tip:

Do not rely on button labels—focus on **physical position**.

- Think in terms of:
- Bottom button
- Right button
- Left button
- Top button

See the images below to compare the location of the inputs of the Nintendo and PlayStation controllers against the Xbox controller.

XBOX BUTTONS AND WHAT THEY DO WHEN USED ON THE NINTENDO SWITCH AND PLAYSTATION

Xbox	Nintendo Switch	PlayStation
A	B	Cross
B	A	Circle
Y	X	Triangle
X	Y	Square

One Handed Gaming

There are many ways to create a one-handed gaming setup, and it is one of the most common requests we receive. Because every person's abilities, goals, and gaming preferences are different, there is no single solution that works for everyone.

One-handed gaming can range from a player using one unaffected hand with typical dexterity to someone who has limited movement and uses only a single finger. The best setup depends on factors such as range of motion, strength, endurance, the types of games being played, and the platform being used. It is often helpful to start by identifying which inputs are difficult to access and which inputs are already available. This can make it easier to select the most effective combination of hardware, software, and controller modifications.

Below are some of the most common one-handed gaming solutions we recommend:

XBOX ADAPTIVE CONTROLLER (XAC) FOR ONE-HANDED PLAY

The XAC is a versatile central hub compatible with Xbox, PC, and mobile platforms.

Benefits:

- **Co-Pilot Mode:** Pair it with a standard controller to allow two devices to function as a single unit.
- **Expandability:** Features 19 3.5mm jacks for external switches and joysticks.
- **Cost Efficiency:** While the base unit is ~$115 CAD, using Makers Making Change (MMC) DIY switches can reduce accessory costs by up to 94% compared to commercial alternatives.

Recommended Add-ons:

- Pair with a mouth joystick like the **LipSync** or **QuadStick** (which features sip-and-puff and gamepad modes) for comprehensive control.

AZERON CONTROLLER

The Azeron is a one-handed keypad and mouse alternative that keeps all necessary inputs within a small reach radius.

Benefits:

- Ergonomic keypad layout
- Significant reduction in hand movement required
- Highly customizable finger placement

Recommended Version:

- The **Azeron Cyro** is the preferred version if the player requires two joysticks combined with keyboard inputs.

Cost:

- Typically around $200 USD.

3D-PRINTED ONE-HANDED CONTROLLER MODS

These are specialized shells that allow standard controllers (Xbox, PlayStation, Nintendo) to be operated with one hand.

Compatibility:

- **PC/Mobile:** Works seamlessly if the game supports standard controller input.
- **Console:** Requires the controller to match the system, or an adapter if using a different brand.
- **Resource:** Use Gaming Readapted's Controller Connect Tool (https://gamingreadapted.com/) to find the correct adapter for your setup.

Link: Test Inputs

Cost:

- Typically <$10 CAD for parts (plus printing/assembly).

CEPHABLE (SOFTWARE)

A free, powerful application for PC and mobile that turns natural movements into game inputs.

Input Methods:

- Voice commands
- Facial expressions
- Head tracking
- Virtual on-screen buttons

Benefits:

- Adds layers of control without the need for additional physical hardware.

PROTEUS CONTROLLER

A premium, fully modular gaming controller designed specifically for accessibility.

Key Features:

- Swappable components to match individual physical needs
- Configurations specifically documented for one-handed users

Cost:

- $400–$470 CAD

3.4 Gamer Profiles

Explore how different individuals set up their gear for success.

3.4.1 Filter Profiles

Player Type

SCI

One-Handed

Type of Access

Alternative Access

Controller Modifications

Software

Platform

Xbox Series X|S

Xbox One

PlayStation 5

Nintendo Switch 1

Nintendo Switch 2

Mobile Gaming

PC Gaming

Filters apply across all profiles below.

3.4.2 Makers Making Change's GAME Player Profile's

These are individuals that have been assisted by Makers Making Change's team directly.

4. GAME Checkpoint Resources

4.1 Game Cheat Sheets

This is a booklet of 10 games that have been documented with basic overviews, links to walkthroughs, and a table of all of the controls needed in the game for PC, Nintendo, Xbox, and PlayStation.

The 10 games are:

- Brawlhalla
- Celeste
- Shadow of the Tomb Raider
- FIFA 23
- DIRT 5
- Mario Kart 8
- Titanfall 2
- Portal 2
- Donut County
- Minecraft